MATH 111 Calculus 1, NUU, FALL 2026

Dr. Salnikova Chekeres **Quiz 2 · Continuity and limits at infinity (1.8, 3.4)**

8

15 minutes. 10 points. No notes, no calculators, no phones. Write your answers in the spaces provided and show all your work. Write "DNE" if a limit does not exist.

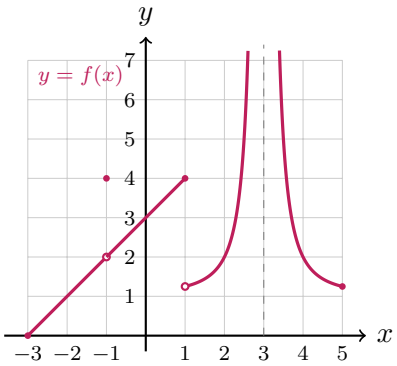
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1	2	3	Total

Problem 1 *Reading a graph.* **[4 points]**

The graph of a function f is shown; the dashed line is a vertical asymptote. Answer each of the following.



(a) $\lim_{x\to -1} f(x) =$ _____

(b) $f(-1) =$ _____

(c) The numbers at which f is discontinuous: _____

(d) At which of them is the discontinuity *removable*, and what value of f there would remove it?

_____

Problem 2 *From the lectures.* **[2 points]**

(Question 9 of Lecture 4, word for word.) Circle one letter, and show the calculation that gives it.

For which c is $g(x) = \begin{cases} cx+1, & x \le 2, \\ x^2 - c, & x > 2 \end{cases}$ continuous everywhere? A student writes four lines; *exactly one is wrong.* Which?

A. Line 1: For continuity at 2 we need $\lim_{x\to 2^-} g(x) = \lim_{x\to 2^+} g(x) = g(2)$.

B. Line 2: $\lim_{x\to 2^-} g(x) = 2c + 1$.

C. Line 3: $\lim_{x\to 2^+} g(x) = c \cdot 2 + 1 = 2c + 1$.

D. Line 4: Therefore g is continuous for every value of c.

Calculation:

Problem 3 *An existence argument.* **[4 points]**

Show that the equation $x^3 - 2x^2 - 4 = 0$ has a solution in the interval $(2, 3)$.

Write the argument in full: name the function, say why it is continuous, give the two values, and state the conclusion.